Environmental Sealing (Dual Wrap Kit) – General Data

1. General

WARNING **SEALANTS MAY CONTAIN TOXIC COMPOUNDS. REFER TO MSDS FOR PROPER PERSONAL PROTECTION EQUIPMENT (PPE) TO USE WHEN WORKING WITH MATERIALS. ALWAYS WEAR GLOVES AND EYE PROTECTION AND AVOID SKIN CONTACT. IF SEALANT OR ANY SEALANT COMPOUND CONTACTS SKIN, CLEAN THOROUGHLY.**

The dual wrap sealing kit is for cold application on harnesses, connectors and/or related connector accessories (backshells, clamps etc.) requiring moisture sealing and abrasion/corrosion protection, for example, fuel gallery.

Note The dual wrap sealing methodology can be applied to installed connector(s), backshell(s) and connector accessories (harness, clamps, shield terminations etc.) all together or to individual components, as required, for environmental sealing.

2. Materials and Equipment

WARNING **USE ONLY FLAME RETARDANT (FR) (FOR EXAMPLE HT3326–5FR–50) ON AIRCRAFT APPLICATIONS DUE TO ITS FLAME RETARDANT CHARACTERISTICS.**

A. Materials

- Stretch Seal Tape, AD89503FR (Flame Retardant)
- Self Fusing Tape, AD59163FR (Flame Retardant)
- Sealant (Potting Compound), HT3326–5FR (Flame Retardant)
- Isopropyl Alcohol (IPA) to Federal Specification TT–I–735, Grade A or B
- Sealant Dispenser (e.g. UG981108–01 or equivalent)

3. Surface Preparation

Prior to installation, ensure the harness bundle and components that will come in contact with the Connector Sealing Kit are free of dirt, oil, grease and other contamination. Remove residue from previously applied adhesives or sealants. Clean harness and components, as required, with isopropyl alcohol.

If possible, un-mate the connector and orient the connector face in a downward position.

4. Preparation of Potting Compound

Remove the injectable polyurethane sealant from the package. Load the cartridge into the sealant dispenser (PN UG981108–01 or equivalent) and follow the package instructions for priming and dispensing the potting compound/sealant HT3326–5FR.

5. Removal/Installation

A. Installation of Connector Sealing Kit

The connector sealing kit consists of Flame Retardant (FR) Stretch Seal Tape (PN AD89503FR), FR Self Fusing Tape (PN AD59163FR) and Polyurethane FR Sealant (PN HT3326–5FR).

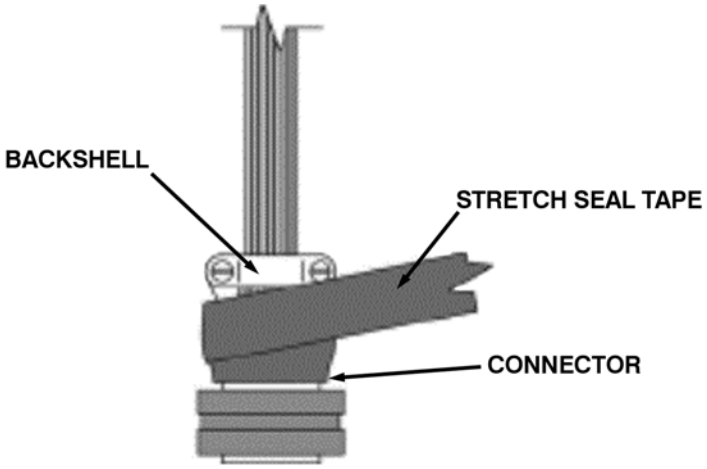
Note HT3326–5FR sealant must be stored and applied at temperatures >55°F. Shelf life of the sealant in the sealed container is six months or as specified in MSDS. The cure time at ambient temperature of 77°F (25°C) is ≤ 4 hours. Any opened cartridge must be used within 24 hours of opening the original packaging. Dispose unused cartridge/sealant as per manufacturer's instructions.

B. Installation of Stretch Seal Tape and Sealant

Install the Stretch Seal Tape (AD89503FR) and polyurethane sealant (HT3326–5FR) as follows:

1) Remove the Stretch Seal Tape from package and unroll a small length of tape. Pull back a small amount of the release film, making sure that the tape remains free of dirt or other contaminants.

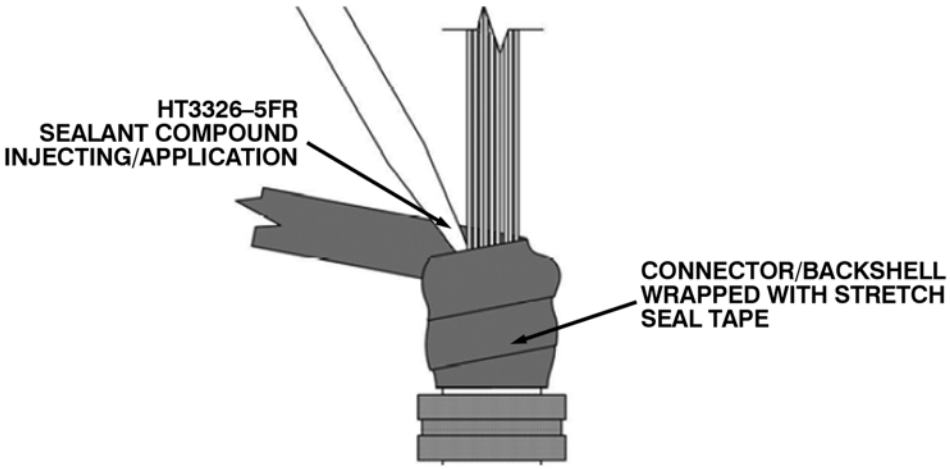
2) Begin wrapping the Stretch Seal Tape around the connector and backshell to be protected, behind the coupling ring (nut) or mating flange/surface. Stretch the tape 25 to 50% to ensure good compression around the connector body and rear accessories (refer to Figure 1). Remove release film as required.



Stretch Seal Tape around Connector and Backshell
Figure 1

3) Continue to wrap the Stretch Seal Tape with a 50% overlap.

4) Stop wrapping the Stretch Seal Tape and hold in place. Immediately inject potting HT3326–5FR (Flame Retardant) compound into the wire bundle behind the connector/backshell rear accessory or through any openings in the rear accessory (refer to Figure 2).

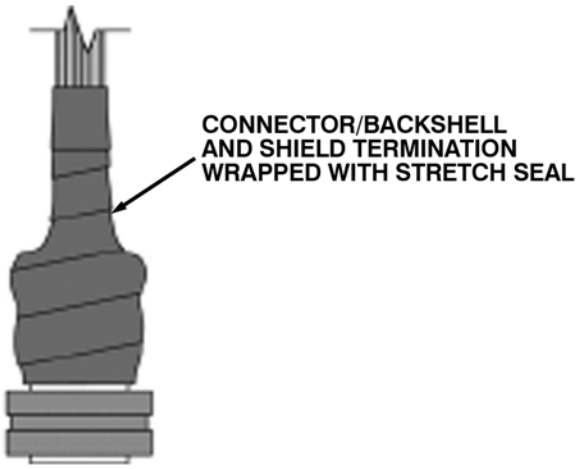


Injecting HT3326–5FR Compound
Figure 2

5) Alternate wrapping the Stretch Seal Tape and injecting the potting compound, until the connector rear accessory is covered, and at least one complete wrap of Stretch Seal Tape has been installed on the harness bundle.

   Note If shield termination is there continue wrapping the Stretch Seal Tape over the shield termination until the back of the connector/backshell is covered, and one complete wrap of Stretch Seal Tape has been installed on the harness bundle.

6) Cut off any excess Stretch Seal Tape. Follow the instructions provided with the potting compound for disposal.



Complete Stretch Seal Tape with Sealing Compound
Figure 3

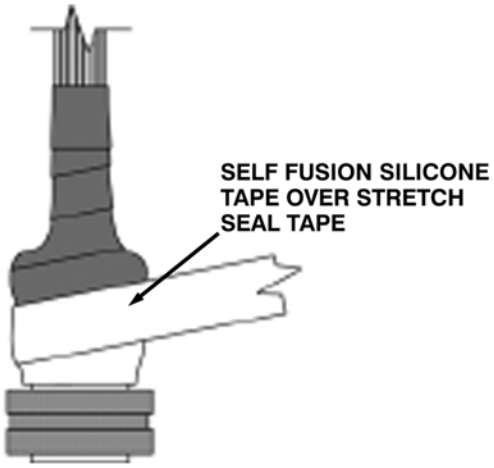
C. Installation of Self Fusing Silicone Tape

Install Self Fusing Silicone Tape (AD59163FR) over the stretch seal tape as follows:

1) Remove the Self Fusing Silicone Tape from the package. Pull back a small amount of the release film, making sure that the tape remains free of dirt or other contaminants.

2) Begin to wrap the connector behind the top of coupling ring (nut) over the stretch seal tape with one complete revolution on itself.

3) After the first full wrap is complete, continue wrapping with a 50% overlap over the top of the Stretch Seal Tape. Use the line in the middle of Self Fusing Silicone Tape as a guide and remove the release film as required (refer to Figure 4).



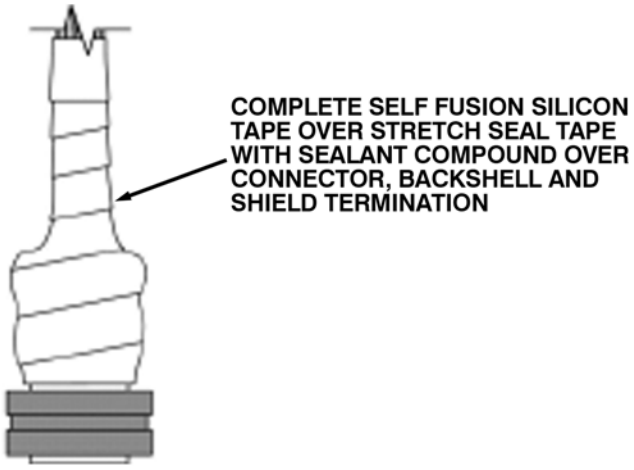
Installation of Self Fusion Silicone Tape
Figure 4

4) Stretch the Self Fusing Silicone Tape 50 to 300% during application to ensure a tight fit when complete. Stretching the tape will aid in the fusing process.

5) Clean any excess potting compound (sealant) that is forced out of the Stretch Seal Tape from the wire bundle.

6) Continue to wrap the Self Fusing Silicone Tape until at least one complete wrap past the end of the stretch seal tape. Make one complete wrap over the top of the last wrap. The first and last wrap must always be a complete wrap directly over the top of the previous wrap.

7) Cut off any excess Self Fusing Silicone Tape.



Complete Environmental Sealing Wrap
Figure 5

D. Environmental Sealing Kit Removal

1) Cut the Self Fusing Silicone Tape with a sharp blade or similar tool, taking care not to cut the harness underneath. Peel the tape away from the harness bundle.

2) Remove the Stretch Seal Tape. Remove the polyurethane sealant by pulling with fingers or other blunt instrument, so as not to damage the wires.

3) Use isopropyl alcohol to remove any remaining residue.

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